EXECUTIVE SUMMARY



VARIA Carbon

AI-Based Soil Carbon Estimation for Carbon MRV

The Challenge

Accurate soil carbon measurement is essential for carbon projects, but conventional laboratory sampling is expensive, time-consuming, and difficult to scale across large landscapes.

Project developers often need a practical way to identify priority areas, estimate baseline soil carbon, and support early-stage project planning before investing in extensive field campaigns.

The Solution

VARIA Carbon is an AI-powered decision-support tool that rapidly estimates **Soil Organic Carbon (SOC%)** using readily available environmental data.

Inputs include:

- NDVI (Satellite Imagery)
- Annual Precipitation
- Mean Annual Temperature
- Elevation
- Clay Content

The platform generates:

- Estimated Soil Organic Carbon (SOC%)
- Carbon stock estimation
- Spatial visualization
- Exportable results for further analysis

VARIA Carbon is designed as a **rapid estimation tool** to complement—not replace—field sampling and laboratory analysis.

Technical Overview

Proof-of-concept model developed and validated using the **EU LUCAS Soil Dataset**.

Performance:

- $R^2 = 0.78$
- MAE = 0.80% SOC
- Low overfitting (train/test performance difference <0.15)

The current version demonstrates the feasibility of AI-assisted soil carbon estimation for large-scale screening and project planning.

Why VARIA Carbon

- AI-assisted estimation in seconds
- No field visit required for initial assessment
- Scalable across large project areas
- Deployable on your own infrastructure
- Complete source code included
- REST API for integration
- Docker-based deployment
- Transparent workflow (no black-box SaaS)

Potential Applications

- Early-stage Carbon MRV
- Baseline soil carbon assessment
- Project area prioritization
- Soil health monitoring
- Sustainable agriculture planning
- Research and education

Current Status

VARIA Carbon is currently available as a working prototype demonstrating the core prediction workflow.

The platform is actively evolving toward a more comprehensive decision-support system for soil carbon and carbon project developers.

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Demo:
https://varia-carbon.streamlit.app